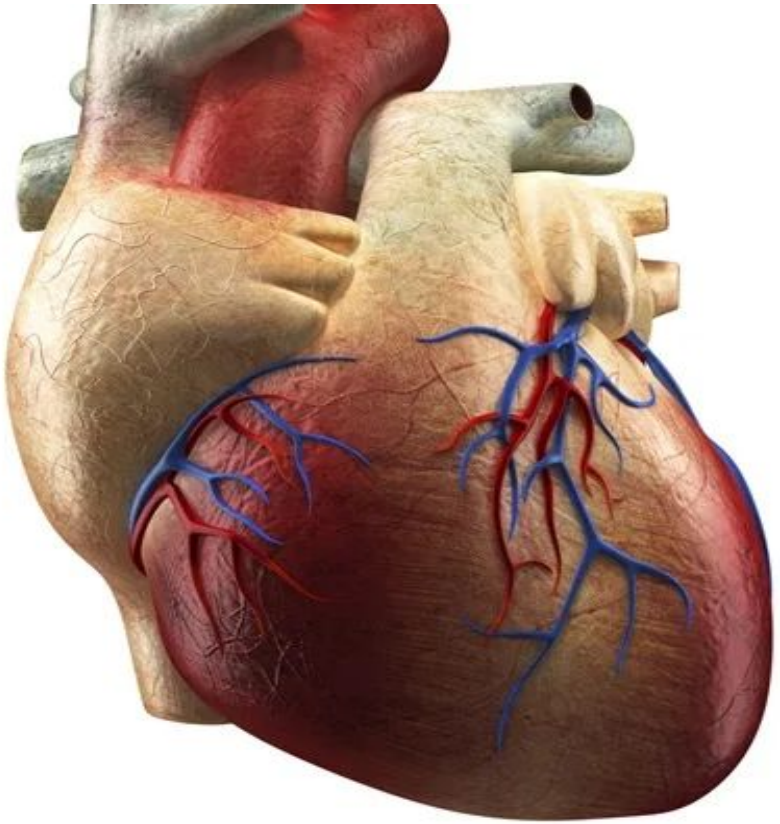
BLOOD SUPPLY OF THE HEART



Learning Objectives:

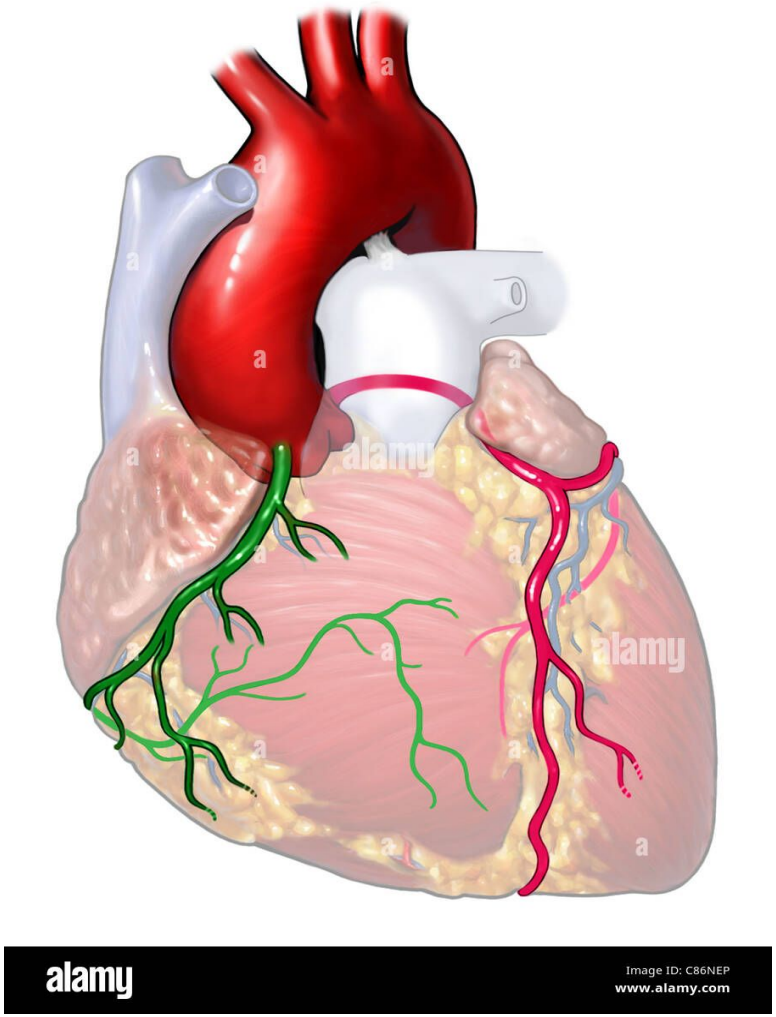
At the end of the lecture the students should be able to:

- **Enlist the major arteries of the heart & their branches.**
- **Describe the course of right & left coronary arteries & their areas of distribution.**
- **Define the coronary sulcus.**
- **Enlist the tributaries of coronary sinus.**
- **Identify the great, middle & small cardiac veins and the major branches of right & left coronary arteries accompanied by them.**

- Blood supply to the heart is known as Coronary Circulation.
- It consists of:

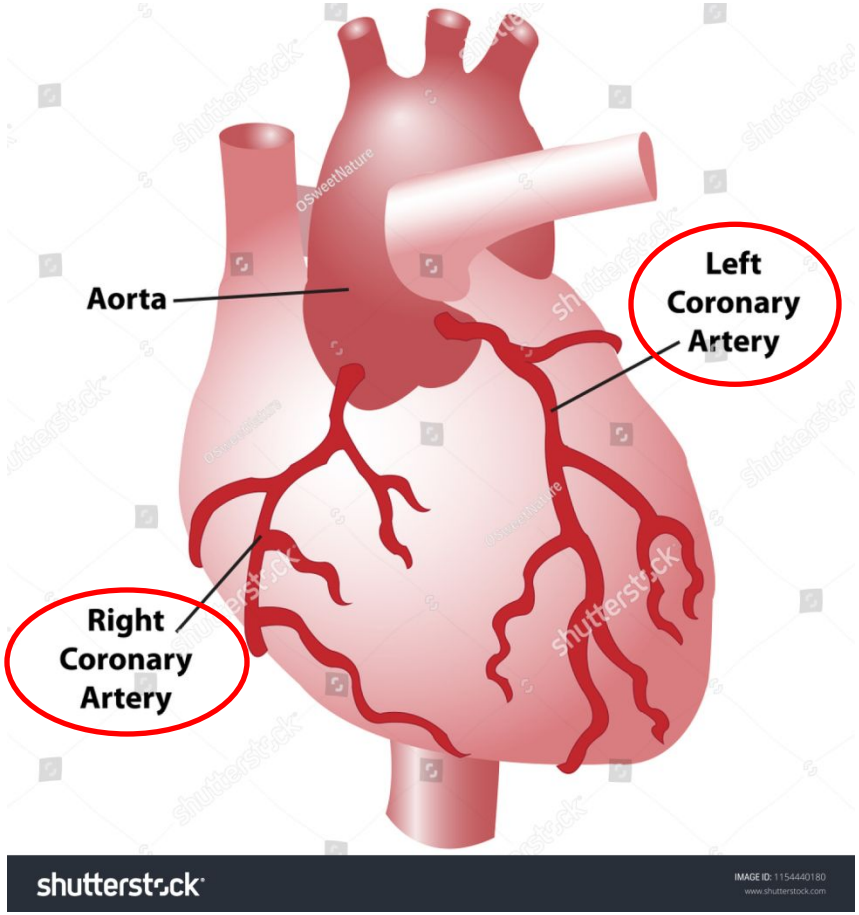
 Arterial supply &

 Venous drainage



Arterial Supply of the Heart:

- The heart is supplied by *two coronary arteries*.
- Both major coronary arteries arise from *ascending aorta*, near the point where left ventricle and pulmonary trunk meet.
- These arteries and their branches supply all parts of the heart.

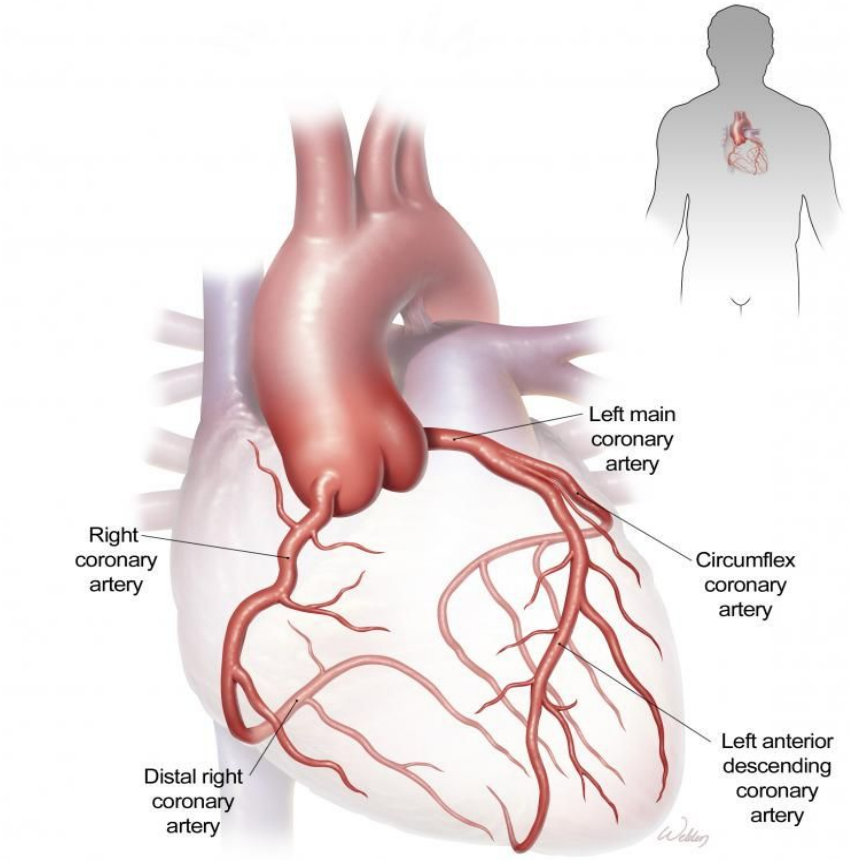


Right Coronary Artery (RCA):

- It is smaller than left coronary artery.
- It arises from anterior aortic sinus of ascending aorta.

Course:

- It emerges on the surface of the heart from root of pulmonary trunk and right auricle.
- It runs downwards in right anterior coronary sulcus.
- Then winds round the inferior border to reach the diaphragmatic surface of the heart.
- Here, it runs in right posterior coronary sulcus and gives posterior inter-ventricular branch which runs in posterior inter-ventricular groove.
- It terminates by anastomosing with the circumflex branch of left coronary artery coming from opposite side in left posterior coronary sulcus.



Branches of RCA:

Large Branches:

i. Posterior inter-ventricular

ii. Marginal

Small Branches:

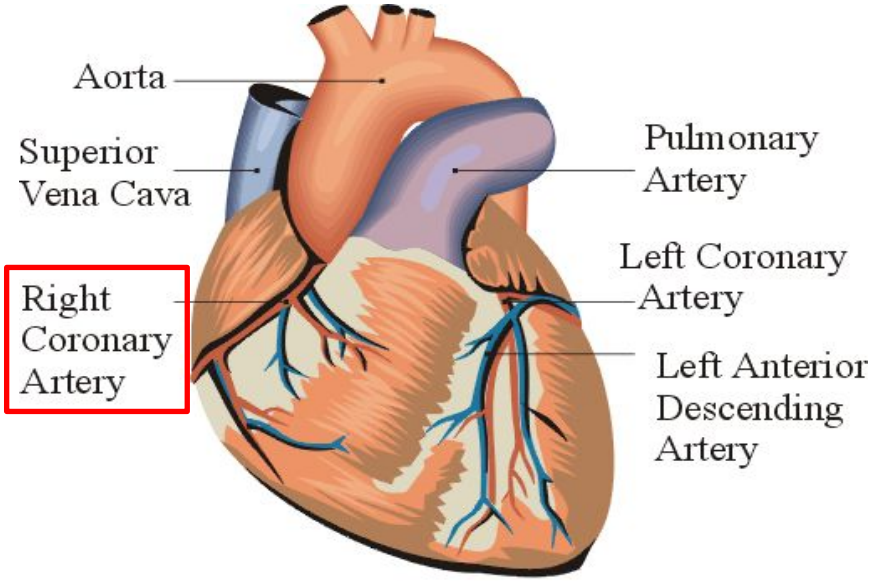
i. Nodal

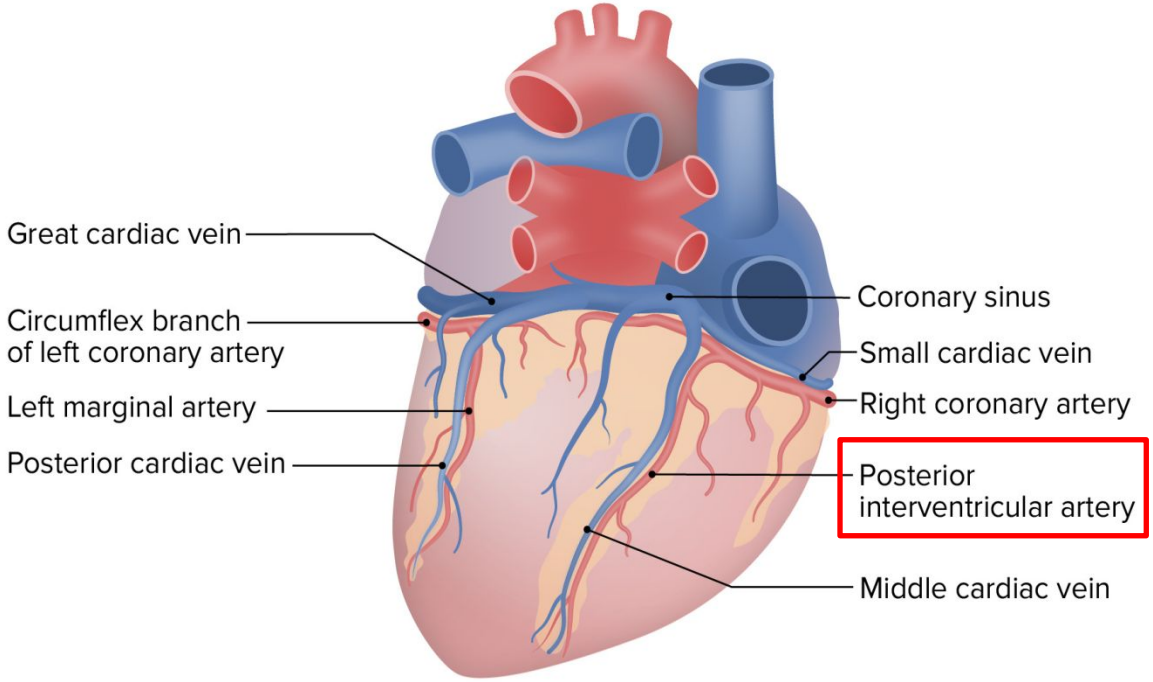
ii. Right atrial

iii. Infundibular

iv. Right ventricular, &

v. Terminal





Area of Distribution of RCA:

- Sino-atrial node (in 60% cases),
- Right atrium,
- Greater part of right ventricle, except the part adjoining anterior inter-ventricular groove,
- A small part of left ventricle adjoining the posterior inter-ventricular groove,
- Posterior part of inter-ventricular septum, &
- Whole of conducting system of the heart except a part of left branch of AV bundle.

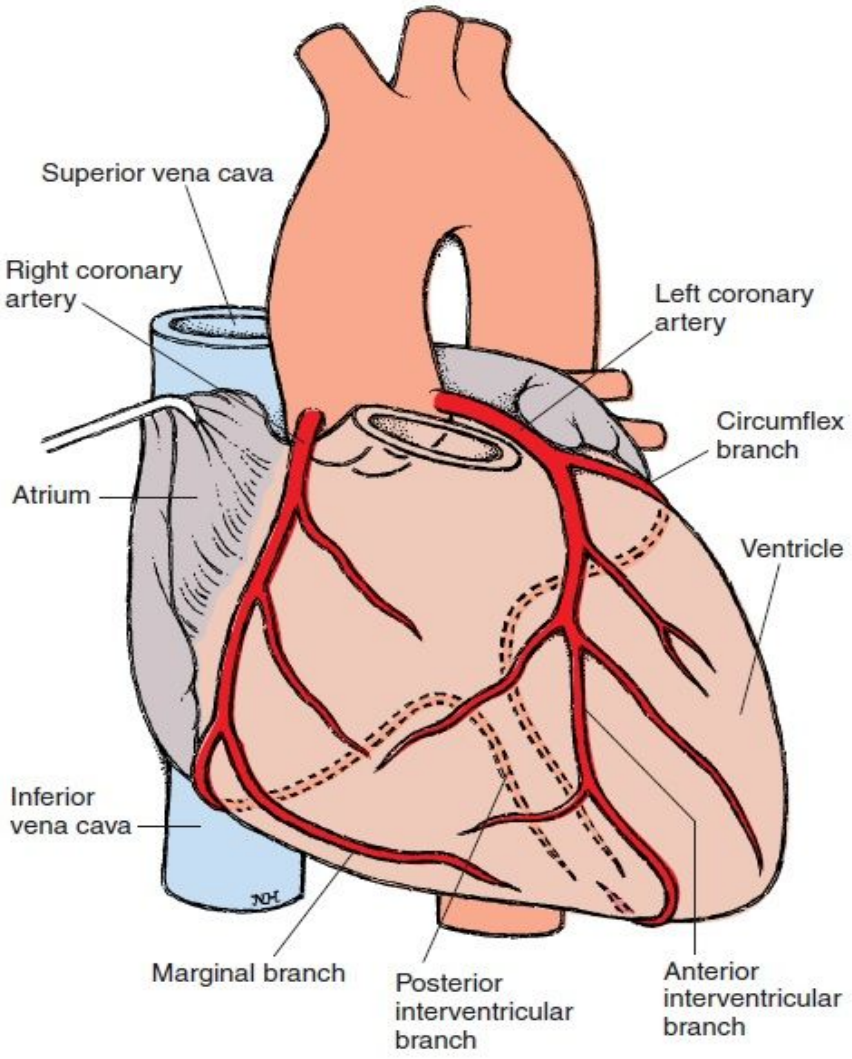


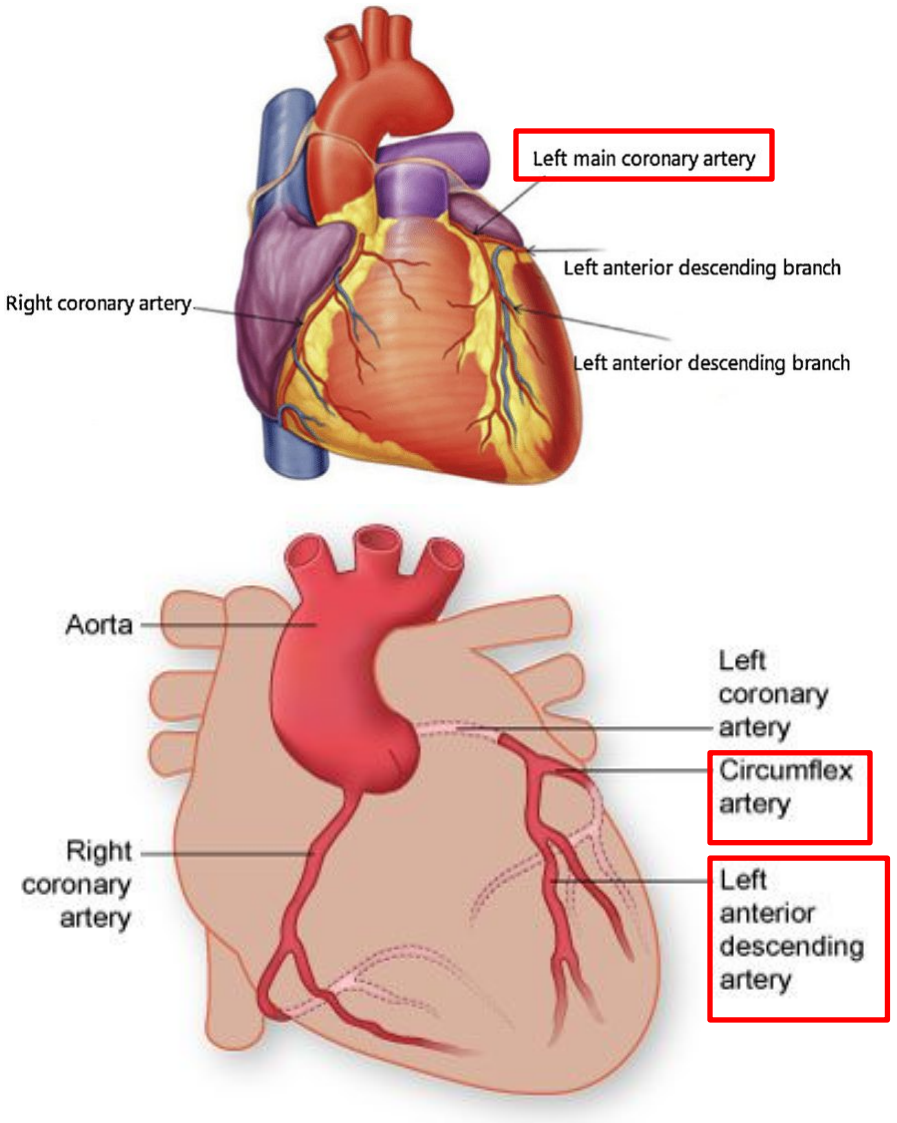
FIGURE 8.14. Coronary Circulation. Anterior view of coronary arteries

Left Coronary Artery (LCA):

- It is larger than right coronary artery.
- It arises from left posterior aortic sinus.

Course:

- It runs forwards and to the left and emerges between pulmonary trunk and the left auricle.
- Here, it gives anterior inter-ventricular artery which runs in anterior inter-ventricular groove.
- After giving off anterior inter-ventricular branch, it runs to the left in left anterior coronary sulcus
- It winds round the left border of the heart and continues in left posterior coronary sulcus as circumflex artery.
- It terminates by anastomosing with the right coronary artery.



Branches & Distribution of LCA:

Large Branches:

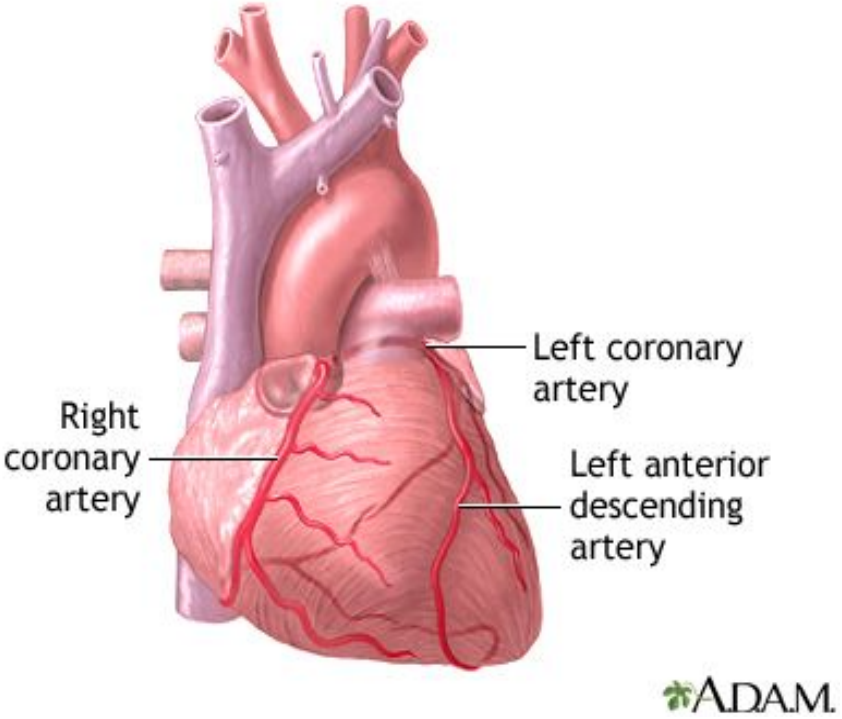
i. Anterior inter-ventricular
ii. Diagonal branch

Small Branches:

i. Left atrial
ii. Pulmonary, &
iii. Terminal

Distribution:

- Left atrium,
- Greater part of left ventricle except, the area adjoining the posterior inter-ventricular groove,
- A small part of right ventricle adjoining anterior inter-ventricular groove,
- Anterior part of inter-ventricular septum, &
- A part of left branch of AV bundle.



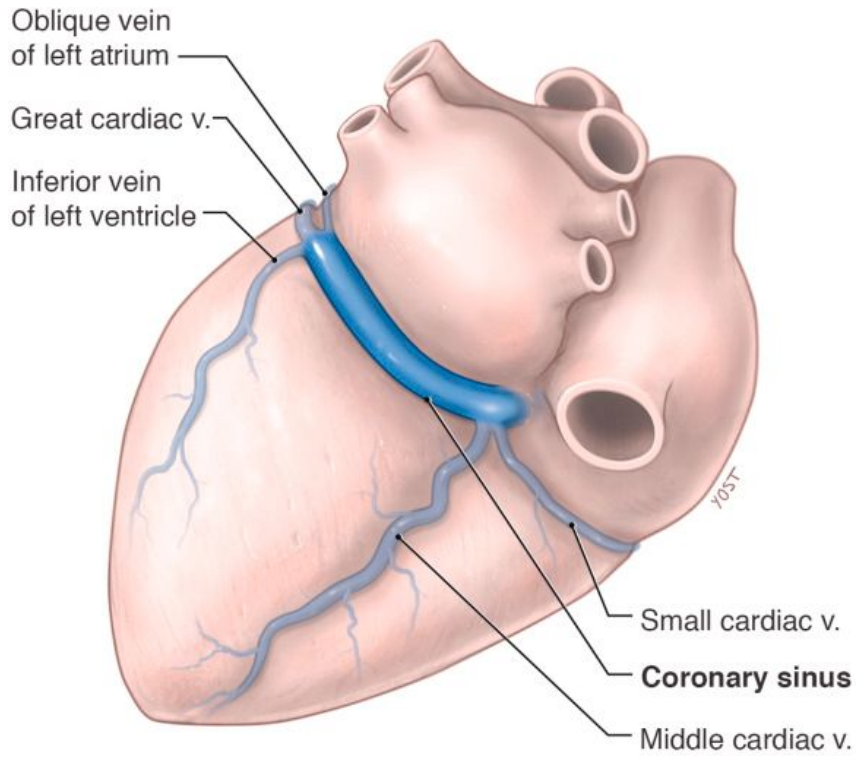
Veins of the Heart:

- The venous drainage of the heart is by means of 3 routes:

A. The Coronary Sinus

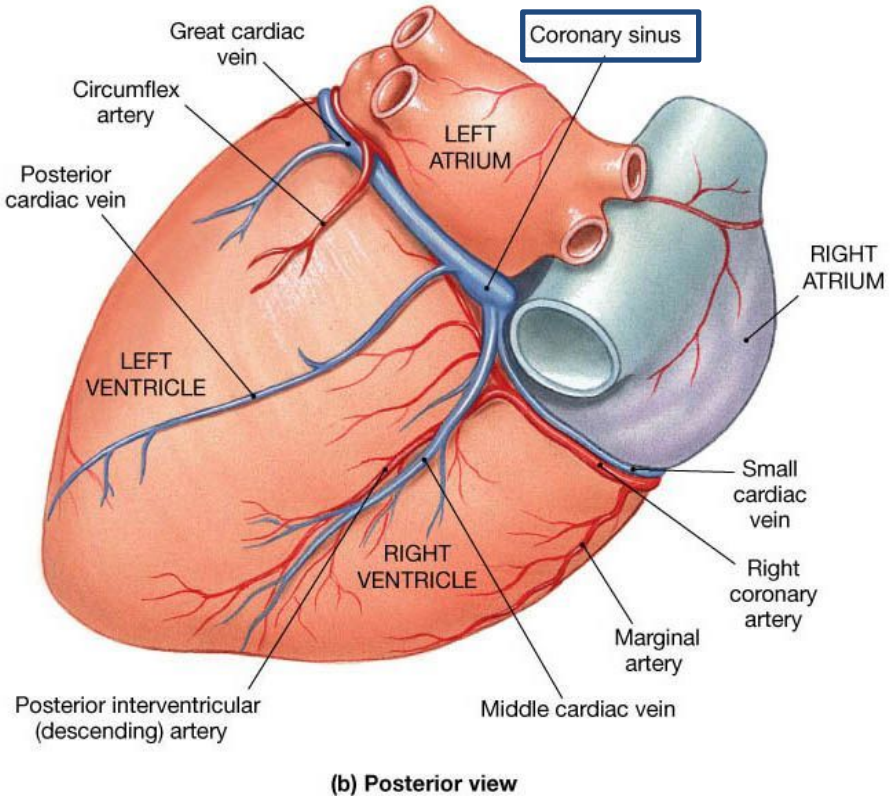
B. Anterior Cardiac Veins

C. Venae Cordis Minimi



A. The Coronary Sinus:

- It is the largest vein of the heart.
- It is about 3 cm long and located in *left posterior coronary sulcus.*
- It ends by opening in the posterior wall of right atrium.



Tributaries of Coronary Sinus:

The coronary sinus has the following tributaries:

i. **Great cardiac vein:** it accompanies *anterior inter-ventricular artery* & then left coronary artery to open in the left end of coronary sinus.

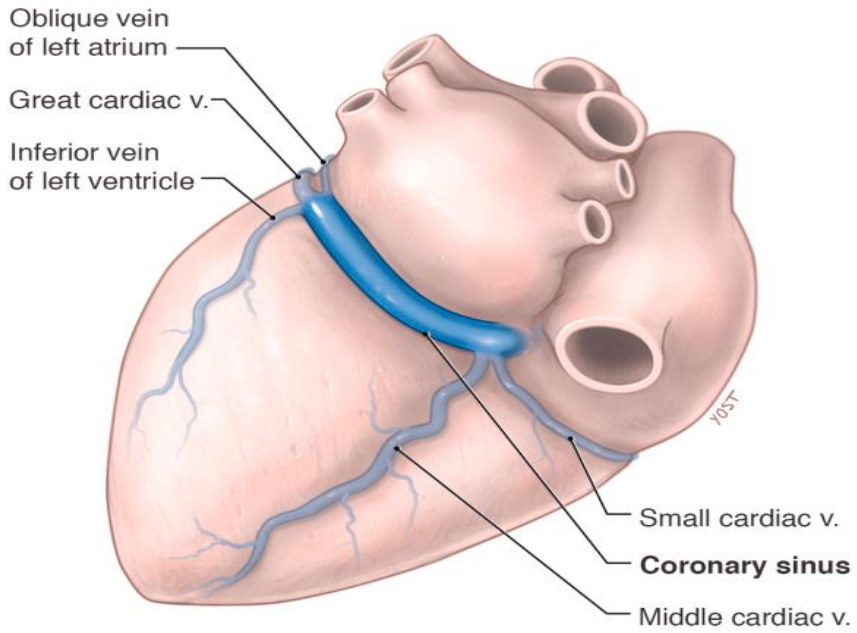
ii. **Middle cardiac vein:** it accompanies *posterior inter-ventricular artery* & opens in the middle of coronary sinus.

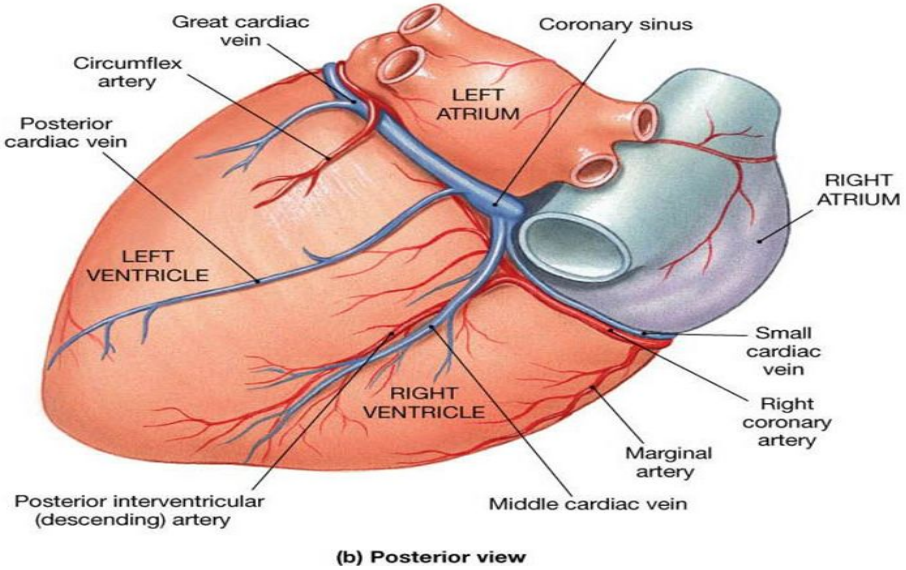
iii. **Small cardiac vein:** it accompanies *right coronary artery* in the right posterior coronary sulcus & opens in the right end of coronary sulcus.

iv. **Posterior vein of left ventricle:** it runs on diaphragmatic surface of left ventricle & drains in the coronary sinus.

v. **Oblique vein of left atrium (of Marshal):** it is a small vein running on the posterior surface of left atrium & ends in coronary sinus.

vi. **Right marginal vein:** it accompanies *marginal branch of RCA* & may either drain into small cardiac vein, or may directly open in the right atrium.





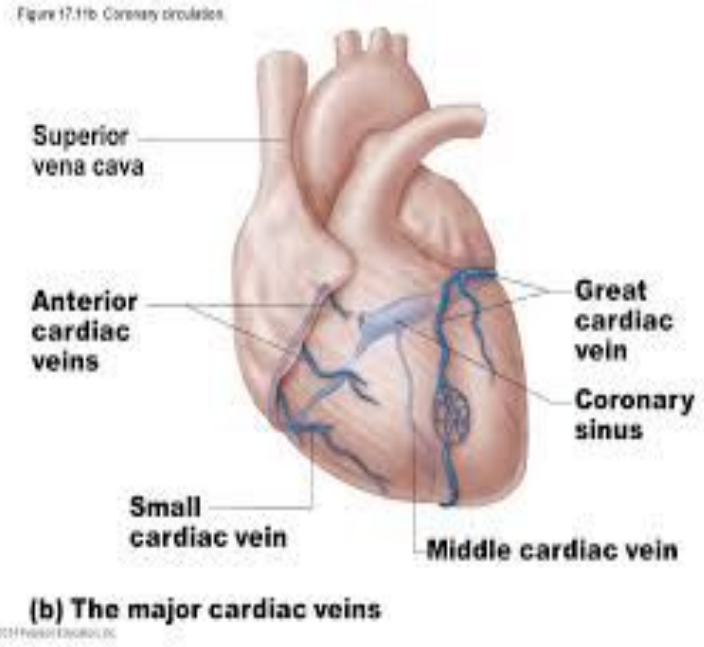
(b) Posterior view

B. Anterior Cardiac Veins:

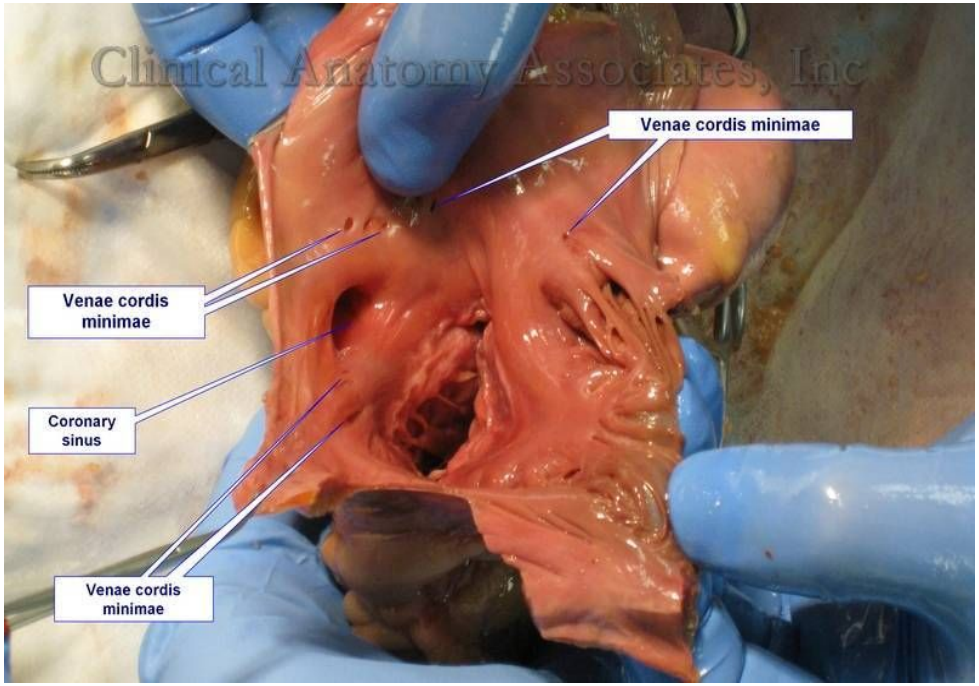
- They are usually a set of 3 to 4 veins which run parallel to each other.
- They open directly into the anterior wall of right atrium.

C. Venae Cordis Minimi:

- Also called Thebesian veins.
- They are numerous small valveless veins present in all 4 chambers of the heart & open directly into their corresponding chamber/cavity.



(b) The major cardiac veins



References:

- **BD CHAURASIA'S HUMAN ANATOMY (Upper limb & Thorax)-Vol. 1**

6th edition

Chapter 18

Pages # 263-267


THANK
YOU